Neural basis of cue reactivity and cue-induced craving: A meta-analysis of neuroimaging studies

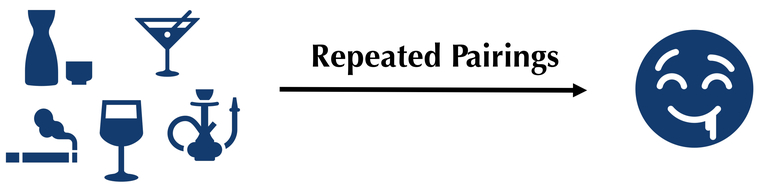
Nicholas R. Harp, R. Ross MacClean, Shosuke Suzuki, Ranjita Poudel, Kaiqi Zhang, & Hedy Kober

Department of Psychiatry, Yale University

Background & Aim

Substance use disorders **(SUDs) are the most prevalent, costly, and deadly of psychiatric disorders**[1].

Drug cues elicit "cue reactivity" – powerful **conditioned physiological responses** in substance users following repeated pairing of cues with drug effects – including **cue-induced craving** (i.e., the desire to use drugs).



Craving is now part of the diagnostic criteria for SUDs and is known to strongly predict drug use and relapse[2].

Previous efforts to consolidate the neuroimaging literature suffer from **low power**, a **focus on specific drugs** (e.g., nicotine/cigarettes), or **suboptimal analytic methods** (e.g., FDR correction, unweighted activations), resulting in **inconsistencies with evidence from pre-clinical and human lesion studies** (e.g., a role for insula).

Study	Ventral Striatum?	Amygdala?	Insula?	Studies?
Chase et al., 2011	Yes	Yes	NO	42
Kühn & Gallinat, 2011	Yes	Yes	NO	29
Engelmann et al., 2012	NO	NO	Yes	11
Schacht et al., 2012	Yes	NO	Yes	28
Tang et al., 2012	Yes	Yes	NO	36
Noori et al., 2016	NO	Yes	Yes	119
Zeng et al., 2021	NO	NO	Yes	21

Here, we provide **the largest meta-analysis to date of the neuroimaging literature examining cue reactivity and cue-induced craving**, including studies with various substance users (e.g., nicotine, opiates, cocaine) and gamblers.

Specific Aims:

Examine cue reactivity and cue-induced craving using state-of-the-art meta-analytic techniques to:

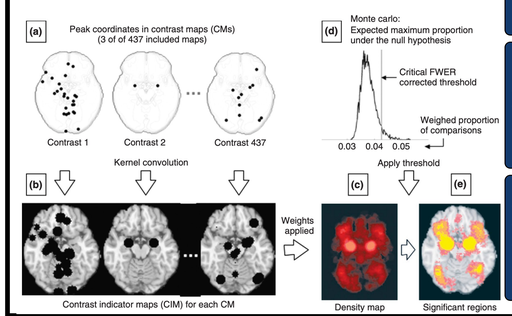
- **Assess across all drugs of abuse/gambling cues**
- **Compare drug types (e.g., alcohol, opiates)**
- **Examine activation specific to subjective cravings**
- **Test use severity, treatment-seeking, and abstinence.**

Method

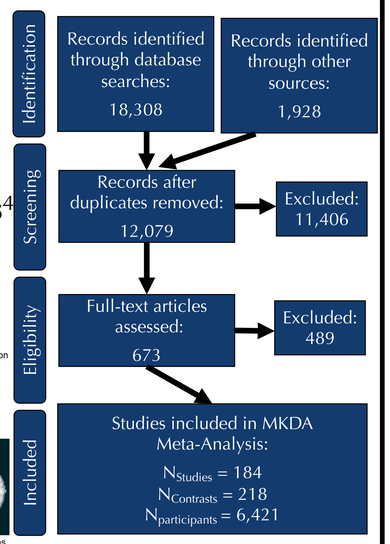
Inclusion Criteria

- Human adult participants;
- Measured cerebral blood flow;
- Used subtraction methodology;
- Contrasted drug > neutral, user > control, or correlation with craving measure;
- Used standard stereotaxic coordinates.

Multi-Level Kernel Density Analysis[4]



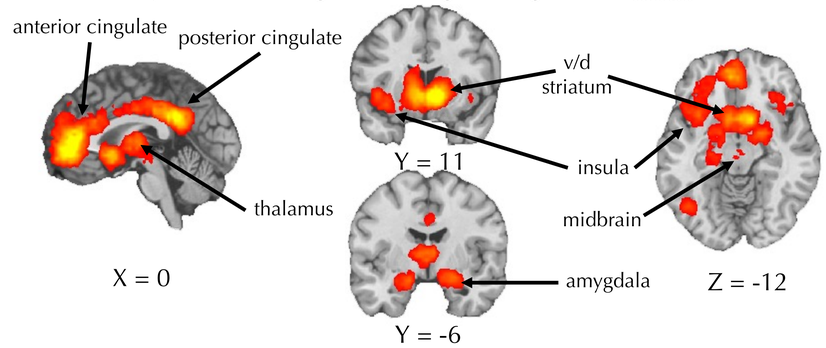
PRISMA Flow Chart[3]



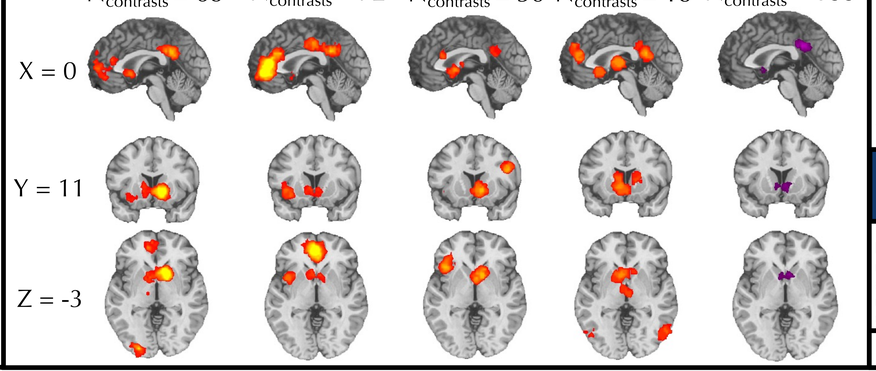
Results

Omnibus contrast

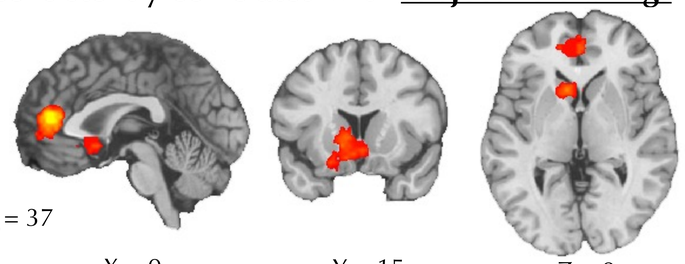
(activity across all drugs of abuse/gambling cues; $N_{contrasts} = 218$)



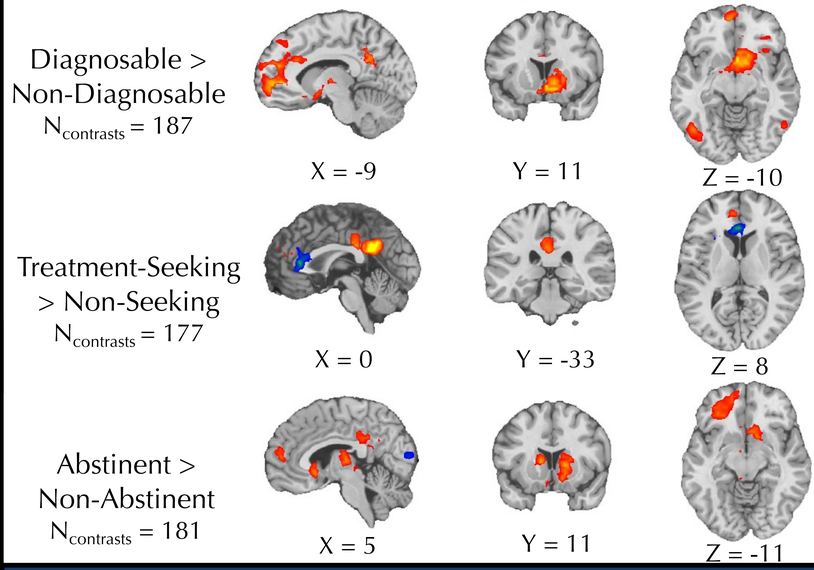
Drug Types



Areas consistently correlated with subjective cravings



Use severity, treatment-seeking, and abstinence



Conclusions

The results **provide convergent evidence with pre-clinical and human lesion studies** (i.e., consistent activation in **ventral striatum, amygdala, insula, and midbrain**), which has been inconsistent/absent from previous meta-analyses.

Areas showing consistent cue reactivity also coincide with a multivariate neuromarker of craving (e.g., ventral striatum, prefrontal cortex, anterior cingulate, and insula)[5].

Together, the results **underscore the brain basis of SUDs** and provide insight for future treatment efforts (e.g., targets for transcranial magnetic stimulation).

References

1. National Institute of Mental Health. (2007). National Comorbidity Survey: Lifetime prevalence estimates. Retrieved from http://www.hcp.med. harvard.edu/ncs/. **2.** Vafaie, N., & Kober, H. (2022). Association of drug cues and craving with drug use and relapse: A systematic review and meta-analysis. *JAMA Psychiatry*, *79*(7), 641-650. **3.** Page et al. (2022). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, *372*, n71. **4.** Kober, H., & Wager, T. D. (2010). Meta-analysis of neuroimaging data. *Wiley Interdisciplinary Review of Cognitive Science*, *1*(2), 293-300. **5.** Koban, L., Wager, T. D., & Kober, H. (2023). A neuromarker for drug and food craving distinguishes drug users from non-users. *Nature Neuroscience*, *26*(2), 316-325.

Funding : R01 DA043690, T32 DA022975 Contact: nicholas.harp@yale.edu